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Creation method of trained model, image generation method, and image processing device

Abstract

In a creation method of a trained model, a reconstructed image (60) obtained by reconstructing three-dimensional X-ray image data (80) is generated. A projection image (61) is generated from a three-dimensional model of an image element (50) by a simulation. The projection image is superimposed on the reconstructed image to generate a superimposed image (67). A trained model (40) is created by performing machine learning using the superimposed image, and the reconstructed image or the projection image.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a creation method of a trained model, an image generation method, and an image processing device.

BACKGROUND ART

(2) In the related art, a method of performing image processing using a trained model is known.

Such a method is disclosed, for example, in PCT International Publication NO. WO2019/138438.

(3) PCT International Publication NO. WO2019/138438 discloses that the conversion is performed on an X-ray image of a region including a specific part of a subject using a trained model to create an image representing the specific part. In PCT International Publication NO. WO2019/138438, a bone part of the subject, a blood vessel into which a contrast medium is injected, and a stent to be placed in a body are shown as an example of the specific part. The trained model is created by performing machine learning using a first digitally reconstructed radiography (DRR) image and a second DRR image, which are reconstructed from CT image data, as a teacher input image and a teacher output image, respectively. By differentiating the image obtained by the conversion using the trained model from an original image, it is possible to generate an image in which the specific part is removed.

CITATION LIST

Patent Literature

(4) [PTL 1] PCT International Publication NO. WO2019/138438

SUMMARY OF INVENTION

Technical Problem

(5) PCT International Publication NO. WO2019/138438 discloses performing, on one specific type of image element (specific part), image processing, such as creating an image representing the bone part as the specific part, creating an image representing a blood vessel into which the contrast medium is injected, and creating an image representing the stent placed in the body. However, in order to improve the visibility of medical images in various usage scenes, it is desired to enable the image processing on various image elements and even on a plurality of image elements, in addition to the image processing limited to one specific image element (specific part).

(6) In addition, PCT International Publication NO. WO2019/138438, since the machine learning is performed using the first DRR image and the second DRR image, which are reconstructed from the CT image data, it is necessary to prepare the CT image data actually including the image element (specific part) to be extracted. In order to efficiently perform the machine learning on various

image elements, it is desired to enable the machine learning without being limited to the use of the CT image data actually including the image element to be extracted, or enable the machine learning even on the image element in which it is difficult to separate and extract only the image element, although it is included in the CT image data.

(7) The present invention has been made to solve the above-described problems, and one object of the present invention is to provide an image generation method and an image processing device capable of performing image processing on various image elements and even on a plurality of image elements, and to provide a creation method of a trained model capable of efficiently performing the creation of the trained model used for such image processing.

Solution to Problem

(8) In order to achieve the above object, a first aspect of the present invention relates to a creation method of a trained model, the method including generating a reconstructed image obtained by reconstructing three-dimensional X-ray image data into a two-dimensional projective image, generating a two-dimensional projection image from a three-dimensional model of an image element, which is an extraction target, by a simulation, superimposing the projection image of the image element on the reconstructed image to generate a superimposed image, and creating a trained model that performs processing of extracting the image element included in an input image, by performing machine learning using the superimposed image as teacher input data and the reconstructed image or the projection image as teacher output data.

(9) A second aspect of the present invention relates to an image generation method including separately extracting a plurality of image elements from an X-ray image using a trained model in which processing of extracting a specific image element from an input image has been learned, and generating a processed image in which image processing is performed on each image element included in the X-ray image, by performing an inter-image arithmetic operation using a plurality of extraction images extracted for the respective image elements, and the X-ray image.

(10) A third aspect of the present invention relates to an image processing device including an image acquisition unit that acquires an X-ray image, an extraction processing unit that separately extracts a plurality of image elements from the X-ray image using a trained model in which processing of extracting a specific image element from an input image has been learned, and an image generation unit that generates a processed image in which image processing is performed on each image element included in the X-ray image, by performing an inter-image arithmetic operation using a plurality of extraction images extracted for the respective image elements, and the X-ray image.

(11) It should be noted that, in the present specification, the “processing of extracting” the image element is a broad concept that includes both generating an image representing the extraction image element by extracting the image element, and generating the X-ray image from which the image element is removed by the extraction. More specifically, the “processing of extracting” the image element includes generating an image of only the image element and generating an image obtained by removing the image element from the original X-ray image. In addition, the “inter-image arithmetic operation” means that one image is generated by performing an arithmetic operation, such as addition, subtraction, multiplication, and division between one image and another image. More specifically, the “inter-image arithmetic operation” means that arithmetic operation processing of a pixel value is performed for each corresponding pixel between a plurality of images to decide the pixel value of the pixel in the image after the arithmetic operation.

Advantageous Effects of Invention

(12) With the creation method of the trained model in the first aspect described above, since the superimposed image obtained by superimposing the reconstructed image obtained by reconstructing the three-dimensional X-ray image data into the two-dimensional projective image on the two-dimensional projection image generated from the three-dimensional model of the image element, which is the extraction target, by a simulation is used as the teacher input data, and the

reconstructed image or the projection image is used as the teacher output data, even in a case in which the three-dimensional X-ray image data does not include the image element, which is the extraction target, it is possible to perform the machine learning using the image element, which is the extraction target, generated by the simulation. That is, teacher data can be prepared without actually preparing a CT image data including the image element to be extracted. In addition, since the projection image of the image element, which is the extraction target, is generated by the simulation, the teacher data can be prepared even for the image element that is included in the CT image data, but is difficult to be separated and extracted. As a result, it is possible to efficiently create the trained model for performing the image processing on various image elements and even on a plurality of image elements.

(13) In addition, with the image generation method in the second aspect described above and the image processing device in the third aspect described above, since the plurality of image elements are separately extracted from the X-ray image using the trained model in which the processing of extracting the specific image element from the input image has been learned, and the processed image is generated by performing the inter-image arithmetic operation using the plurality of extraction images extracted for the respective image elements, and the X-ray image, it is possible to freely add or subtract each extraction image from the X-ray image in accordance with the type of the extraction image element after various image elements are separately extracted from the input X-ray image as the extraction images. As a result, the image processing can be performed on various image elements and even on a plurality of image elements.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a block diagram showing an image processing device according to an embodiment.
- (2) FIG. 2 is a diagram showing an example of an X-ray imaging device.
- (3) FIG. 3 is a diagram for describing machine learning with respect to a learning model, and a trained model.
- (4) FIG. 4 is a diagram showing an example of an image element.
- (5) FIG. 5 is a diagram showing a first example of the extraction of the image element by the trained model.
- (6) FIG. 6 is a diagram showing a second example of the extraction of the image element by the trained model and the generation of a processed image.
- (7) FIG. 7 is a diagram showing an example of performing image processing on an extraction image, which is different from FIG. 6.
- (8) FIG. 8 is a flowchart for describing an image generation method according to the embodiment.
- (9) FIG. 9 is a flowchart for describing a creation method of the trained model according to the embodiment.
- (10) FIG. 10 is a diagram for describing the generation of a superimposed image used for the machine learning.
- (11) FIG. 11 is a diagram for describing a generation method of a reconstructed image.
- (12) FIG. 12 is a diagram showing an example of teacher input data and teacher output data in a case in which the image element is a bone.
- (13) FIG. 13 is a diagram showing an example of the processed image in a case in which the image element is the bone.
- (14) FIG. 14 is a diagram showing an example of the teacher input data and the teacher output data in a case in which the image element is a device.
- (15) FIG. 15 is a diagram showing an example of the processed image in a case in which the image element is the device.

- (16) FIG. **16** is a diagram showing variations (A) to (I) of a projection image of the device generated from a three-dimensional model.
- (17) FIG. **17** is a diagram showing an example of the teacher input data and the teacher output data in a case in which the image element is a noise.
- (18) FIG. **18** is a diagram showing an example of the processed image in a case in which the image element is the noise.
- (19) FIG. **19** is a diagram showing an example of the teacher input data and the teacher output data in a case in which the image element is a blood vessel.
- (20) FIG. **20** is a diagram showing an example of the processed image in a case in which the image element is the blood vessel.
- (21) FIG. **21** is a diagram showing an example of the teacher input data and the teacher output data in a case in which the image element is clothing.
- (22) FIG. **22** is a diagram showing an example of the processed image in a case in which the image element is the clothing.
- (23) FIG. **23** is a diagram showing an example of the teacher input data and the teacher output data in a case in which the image element is a scattered ray component of X-rays.
- (24) FIG. **24** is a diagram showing an example of the processed image in a case in which the image element is the scattered ray component of the X-rays.
- (25) FIG. **25** is a diagram showing a projection angle range of the X-rays in a first direction (A) and a second direction (B) in a Monte Carlo simulation.
- (26) FIG. **26** is a diagram showing a first example of an energy spectrum of the X-rays in the Monte Carlo simulation.
- (27) FIG. **27** is a diagram showing a second example of the energy spectrum of the X-rays in the Monte Carlo simulation.
- (28) FIG. **28** is a diagram showing an example of a collimator image used for the teacher input data and the teacher output data.
- (29) FIG. **29** is a diagram showing an example of extracting the image elements from a plurality of trained models.

DESCRIPTION OF EMBODIMENTS

- (30) In the following, an embodiment embodying the present invention will be described with reference to the drawings.
- (31) With reference to FIGS. **1** to **28**, a configuration of an image processing device **100** according to the embodiment, an image generation method according to the embodiment, and a creation method of a trained model according to the embodiment will be described.
- (32) (Configuration of Image Processing Device)
- (33) First, the configuration of the image processing device **100** will be described with reference to FIG. **1**.
- (34) The image processing device **100** is configured to extract an image element **50** included in an X-ray image **201** using a trained model **40** created through machine learning, and perform image processing of the X-ray image **201** using the extraction image element **50**. The image processing device **100** generates the X-ray image **201** captured by an X-ray imaging device **200** as an input, and a processed image **22** in which the image processing is performed on each image element **50** included in the X-ray image **201** as an output.
- (35) As shown in FIG. **1**, the image processing device **100** includes an image acquisition unit **10**, an extraction processing unit **20**, and an image generation unit **30**. The image processing device **100** is configured by a computer including one or a plurality of processors **101**, such as a central processing unit (CPU) and a graphics processing unit (GPU), and one or a plurality of storage units **102**, such as a hard disk drive (HDD) and a solid state drive (SSD). The image processing device **100** is connected to a display device **103**.
- (36) The image acquisition unit **10** is configured to acquire the X-ray image **201**. The image

acquisition unit **10** is configured by, for example, an interface for communicably connecting an external device and the image processing device **100**. The image acquisition unit **10** may include a communication interface, such as a local area network (LAN) as an interface. The image acquisition unit **10** may include an input/output interface, such as HDMI (registered trademark), a display port, and a USB port. The image acquisition unit **10** can acquire the X-ray image **201** from the X-ray imaging device **200** or from a server device connected via a network by communication. (37) The X-ray image **201** is a medical image obtained by imaging a patient or a subject by the X-ray imaging device **200**. The X-ray image **201** may be any a still image or a video image. The video image is a collection of still images captured at a predetermined frame rate. The X-ray image **201** is a two-dimensional image. The X-ray image **201** may be various images captured by simple X-ray imaging, fluoroscopy, angiography, and the like.

(38) The extraction processing unit **20** is configured to separately extract a plurality of image elements **50** from the X-ray image **201** using the trained model **40** stored in the storage unit **102**. The trained model **40** is a trained model in which processing of extracting a specific image element **50** from an input image has been learned.

(39) The image element **50** is an image part or image information constituting the X-ray image **201**, and is defined as a group of the same or similar types. The image element **50** may be a part of a human body that is anatomically classified. Such an image element **50** is a biological tissue, such as a bone or a blood vessel. The image element **50** may be an object introduced or placed in the body of the subject in surgery or the like. Such an image element **50** may be, for example, a device, such as a catheter, a guidewire, a stent, a surgical instrument, or fixture, which is introduced into the body. The image element **50** may be a noise, an artifact, a scattered ray component of X-rays, or the like generated during imaging processing in the X-ray imaging. The image element **50** may be clothing worn by the subject and reflected at the time of the imaging. It should be noted that the clothing is a concept including clothes, ornaments, and other attachments. For example, a button, a fastener, an accessory, a metal part, and the like of the clothing are reflected in the X-ray image **201**.

(40) The trained model **40** is created in advance through the machine learning in which the processing of extracting the specific image element **50** from the input image is learned. The extraction processing unit **20** extracts the image element **50** using one or a plurality of trained models **40**. As a result, the extraction processing unit **20** generates a plurality of extraction images **21** obtained by extracting separate image elements **50** from the X-ray image **201**. For example, a first extraction image **21** includes a first image element **50**, and a second extraction image **21** includes a second image element **50** different from the first image element **50**. The creation method of the trained model **40** will be described below.

(41) The image generation unit **30** is configured to generate the processed image **22** in which the image processing is performed on each image element **50** included in the X-ray image **201**, by performing inter-image arithmetic operation using the plurality of extraction images **21** extracted for the respective image elements **50**, and the X-ray image **201**. The image processing includes, for example, enhancement processing of the image element **50** or removal processing of the image element **50**. It should be noted that the enhancement processing is processing of relatively increasing a pixel value of a pixel belonging to the image element **50**. The removal processing is processing of relatively lowering the pixel value of the pixel belonging to the image element **50**. The removal processing includes reducing the visibility by partial removal, in addition to completely removing the image from the image. In addition, the enhancement processing may be, for example, edge enhancement processing. The removal processing may be noise removal processing.

(42) The inter-image arithmetic operation is to decide the pixel value of the corresponding pixel in the processed image **22** by performing the arithmetic operation of the pixel value for each corresponding pixel between the plurality of extraction images **21** and the X-ray image **201**. A

content of the arithmetic operation is not particularly limited, but may be, for example, four arithmetic operations of addition, subtraction, multiplication, and division. In the present embodiment, the inter-image arithmetic operation includes weighting addition or weighting subtraction of the individual extraction image **21** with respect to the X-ray image **201**. By performing the weighting addition of the extraction image **21** with respect to the X-ray image **201**, the enhancement processing of the image element **50** included in the X-ray image **201** can be performed. By performing the weighting subtraction of the extraction image **21** with respect to the X-ray image **201**, the removal processing of the image element **50** included in the X-ray image **201** can be performed. By adjusting a weight value, a degree of enhancement or a degree of removal of the image element **50** can be optimized.

(43) In the example of FIG. **1**, the processor **101** functions as the extraction processing unit **20** and the image generation unit **30** by executing a program (not shown) stored in the storage unit **102**. That is, in the example of FIG. **1**, the extraction processing unit **20** and the image generation unit **30** are realized as functional blocks of the processor **101**. The extraction processing unit **20** and the image generation unit **30** may be configured as individual hardware.

(44) It should be noted that the individual hardware includes the extraction processing unit **20** and the image generation unit **30** configured by separate processors. The individual hardware means that a plurality of computers (PCs) in the image processing device **100**, and a computer (PC) that functions as the extraction processing unit and a computer (PC) that functions as the image generation unit are separately provided.

(45) The image processing device **100** displays, on the display device **103**, the processed image **22** generated by the image generation unit **30**. For example, the image processing device **100** transmits the generated processed image **22** to the server device via the network. For example, the image processing device **100** records the generated processed image **22** in the storage unit **102**.

(46) FIG. **2** shows a configuration example of the X-ray imaging device **200**. FIG. **2** shows an example of a blood vessel X-ray imaging device that can perform blood vessel fluoroscopy. The X-ray imaging device **200** includes a top plate **210**, an X-ray irradiation unit **220**, and an X-ray detector **230**. The top plate **210** is configured to support a subject **1** (person). The X-ray irradiation unit **220** includes an X-ray source, such as an X-ray tube, and is configured to emit X-rays toward the X-ray detector **230**. The X-ray detector **230** is configured by, for example, a flat panel detector (FPD), and is configured to detect the X-rays emitted from the X-ray irradiation unit **220** and transmitted through the subject **1**.

(47) In the example of FIG. **2**, the X-ray irradiation unit **220** and the X-ray detector **230** are held by a C-arm **240**. The C-arm **240** is movable in a first direction **250** along an arm portion having an arcuate shape and is rotatable in a second direction **252** around a rotation axis **251**. As a result, the X-ray imaging device **200** can change the projection direction of the X-rays from the X-ray irradiation unit **220** toward the X-ray detector **230** to the first direction **250** and the second direction **252**, respectively, by a predetermined angle range.

(48) (Trained Model)

(49) As shown in FIG. **1**, the processing of extracting the image element **50** included in the X-ray image **201** is performed by the trained model **40** created through the machine learning. As shown in FIG. **3**, the trained model **40** extracts a pre-learned image element **50** from the input image, and outputs the extraction image **21** displaying only the extraction image element **50**.

(50) In the present embodiment, the trained model **40** is created in advance through the machine learning using a reconstructed image **60** obtained by reconstructing three-dimensional image data into a two-dimensional projective image, and a projection image **61** created from a three-dimensional model of the image element **50** by a simulation.

(51) As a machine learning method, any method, such as fully convolutional neural networks (fully convolutional networks; FCN), a neural network, a support vector machine (SVM), or boosting, can be used. For a learning model LM (trained model **40**) according to the present embodiment, it

is preferable to use a convolutional neural network, and more preferably the fully convolutional neural networks. Such a learning model LM (trained model **40**) includes an input layer **41** into which an image is input, a convolutional layer **42**, and an output layer **43**.

(52) In order to create the trained model **40**, the machine learning is performed using training data **66** including teacher input data **64** and teacher output data **65**. The teacher input data **64** and the teacher output data **65** included in one training data **66** have a relationship between the data before extraction and the data after extraction for the same image element **50**.

(53) The machine learning is performed for each image element of the plurality of image elements **50** to be extracted. That is, the training data **66** is prepared for each image element **50** to be extracted.

(54) As shown in FIG. 4, the plurality of image elements **50** include a first element **51**, which is the biological tissue, and a second element **52**, which is a non-biological tissue. In addition, the plurality of image elements **50** include at least a plurality of image elements of a bone **53**, a blood vessel **54**, a device **55** introduced into the body, clothing **56**, a noise **57**, and a scattered ray component **58** of the X-rays. Among these, the bone **53** and the blood vessel **54** correspond to the first element **51**. The first element **51** may include the biological tissue other than the bone **53** and the blood vessel **54**. Among these, the device **55** introduced into the body, the clothing **56**, the noise **57**, and the scattered ray component **58** of the X-rays correspond to the second element **52**. The second element **52** may include the image element other than the device **55**, the clothing **56**, the noise **57** and the scattered ray component **58**.

(55) In the example shown in FIG. 5, one trained model **40** is configured to separately extract the plurality of image elements **50**. The trained model **40** has one input channel and a plurality (N) of output channels. N is an integer of 2 or more. In a case in which the X-ray image **201** is input to the input channel, the trained model **40** separately extracts N image elements **50**. The trained model **40** outputs the extracted first to N-th image elements **50** as a first extraction image **21-1** to an N-th extraction image **21-N** from N output channels, respectively.

(56) As an example different from FIG. 5, in the example of FIG. 6, the trained model **40** has one input channel and N+1 output channels. In a case in which the X-ray image **201** is input to the input channel, the trained model **40** extracts a plurality (N) of image elements **50** from the input image without duplication. "Without duplication" means that the image information extracted in any of the extraction images (for example, the first extraction image **21-1**) is not included in other extraction images (for example, the second extraction image **21-2** to the N-th extraction image **21-N**). The trained model **40** outputs the extracted first to N-th image elements as the first extraction image **21-1** to the N-th extraction image **21-N**, respectively. Moreover, the trained model **40** outputs a residual image element **59** remaining after extraction as an N+1-th extraction image **21x** from an N+1-th output channel.

(57) In this case, the first extraction image **21-1** to the N-th extraction image **21-N** do not include the same image information. Moreover, among the input X-ray images **201**, the image information remaining without being extracted is included in the N+1-th extraction image **21x**. Therefore, in a case in which the first extraction image **21-1** to the N-th extraction image **21-N** and the N+1-th extraction image **21x** are added, the original X-ray image **201** is returned.

(58) As described above, in the example of FIG. 6, the trained model **40** is configured to extract the plurality of image elements **50** from the input image without duplication, and output the extracted plurality of image elements **50** and the residual image element **59** remaining after extraction, respectively. As a result, the total of the image information extracted by the extraction processing is neither increased nor decreased as compared with the input image.

(59) As shown in FIG. 6, the image generation unit **30** (see FIG. 1) generates the processed image **22** by performing, on the input X-ray image **201**, the inter-image arithmetic operation of addition or subtraction by adding a weight coefficient **23** to the first extraction image **21-1** to the N-th extraction image **21-N**. The N+1-th extraction image **21x** representing the residual image element

59 does not need to be used for the image processing. As the weight coefficient **23**, coefficients **w1** to **wn** are individually set corresponding to the first extraction image **21-1** to the N-th extraction image **21-N**, respectively. The weight coefficient **23** may be, for example, a fixed value set in advance in the storage unit **102**. It should be noted that a plurality of types of weight coefficients may be set in accordance with the intended use of the processed image **22** and the like. For example, a first setting value of the weight coefficient **23** is used for the first extraction image **21-1** in a processing mode A, and a second setting value of the weight coefficient **23** is used for the first extraction image **21-1** in a processing mode B. In addition, the weight coefficient **23** may be set to any value in accordance with the operation input of a user.

(60) As shown in FIG. 7, the image processing may be performed on each of the extraction images **21** before performing the inter-image arithmetic operation between each extraction image **21** and the X-ray image **201**. In the example of FIG. 7, the image generation unit **30** is configured to separately perform the image processing on a part or all of the plurality of extraction images **21**. The processed image **22** is generated by the inter-image arithmetic operation between the plurality of extraction images **21** after the image processing and the X-ray image **201**.

(61) For example, in FIG. 7, the image generation unit **30** performs first image processing **25-1** on the first extraction image **21-1**, performs second image processing **25-2** on the second extraction image **21-2**, . . . , and performs N-th image processing **25-N** on the N-th extraction image **21-N**. In some cases, it is not necessary to perform the image processing depending on the image element **50**, and thus the image processing may be performed only on a part of the extraction images **21**.

(62) The image processing performed on the individual extraction image is not particularly limited, and may be image correction processing or image interpolation processing, for example. The image correction processing may include the edge enhancement processing and the noise removal processing. The image correction processing may be, for example, contrast adjustment, line enhancement processing, and smoothing processing. The image interpolation processing is processing of interpolating an interrupted part of the image element **50**, which appears to be interrupted in the middle because it is difficult to be captured in the X-ray image **201**, such as the guidewire or the catheter. In the contrast adjustment, the line enhancement processing, or the like, an appropriate parameter differs for each image element **50**, and it is difficult to process all the image elements **50** at once, but the image processing is performed on the individual extraction image **21**, so that the optimum image processing is performed on each image element **50**.

(63) (Image Generation Method)

(64) Next, the image generation method according to the present embodiment will be described with reference to FIG. 8. The image generation method can be performed by the image processing device **100**. The image generation method according to the present embodiment includes at least following step S2 and step S5 shown in FIG. 8.

(65) (S2) The plurality of image elements **50** are separately extracted from the X-ray image **201** using the trained model **40** in which the processing of extracting the specific image element **50** from the input image has been learned.

(66) (S5) The processed image **22** in which the image processing is performed on each image element **50** included in the X-ray image **201** is generated by performing the inter-image arithmetic operation using the plurality of extraction images **21** extracted for the respective image elements **50**, and the X-ray image **201**.

(67) In addition, the image generation method according to the present embodiment may further include steps S1, S3, S4, and S6 shown in FIG. 8.

(68) In step S1, the X-ray image **201** is acquired. Specifically, the image acquisition unit **10** (see FIG. 1) acquires the X-ray image **201** captured by the X-ray imaging device **200** shown in FIG. 2 by, for example, communication with the X-ray imaging device **200** or the server device.

(69) In step S2, the extraction processing unit **20** separately extracts the plurality of image elements **50** from the X-ray image **201** using the trained model **40**. The extraction processing unit **20** inputs

the X-ray image **201** acquired in step **S1** to the trained model **40**. As a result, the trained model **40** outputs the first extraction image **21-1** to the N-th extraction image **21-N** as shown in FIG. 5 or FIG. 6.

(70) In step **S3**, as shown in FIG. 7, the image processing may be performed on a part or all of the extracted plurality of extraction images **21**. In this case, the image generation unit **30** performs preset image processing on the extraction image **21**, which is a target of the image processing, with a predetermined parameter. Whether or not to perform the image processing may be decided in accordance with the input from the user. Whether or not to perform the image processing may be decided in accordance with an image quality of the extraction image **21**. Step **S3** does not have to be performed.

(71) In step **S4**, the image generation unit **30** acquires an arithmetic operation parameter for each extraction image **21**. The arithmetic operation parameters include, for example, a setting value of the weight coefficient **23** and a setting value of an arithmetic operation method. The setting value of the arithmetic operation method represents whether to perform the weighting addition (that is, the enhancement processing of the image element **50**) or to perform the weighting subtraction (that is, the removal processing of the image element **50**) for the extraction image **21**, which is the target. The setting value of the arithmetic operation method and the setting value of the weight coefficient **23** are preset in the storage unit **102** for each type of the image element **50** to be extracted.

(72) In step **S5**, the image generation unit **30** performs the inter-image arithmetic operation using the plurality of extraction images **21** extracted for the respective image elements **50**, and the X-ray image **201**. The image generation unit **30** performs the inter-image arithmetic operation in accordance with the parameters acquired in step **S4**. The image generation unit **30** multiplies each of the first extraction image **21-1** to the N-th extraction image **21-N** by the corresponding weight coefficient **23**, and performs the inter-image arithmetic operation on the X-ray image **201** by the corresponding arithmetic operation method. As a result, the X-ray image **201** subjected to the weighting addition or the weighting subtraction by each extraction image **21** is generated as the processed image **22**. As a result, the image generation unit **30** generates the processed image **22** (see FIG. 6 or 7) in which the image processing, which is the enhancement processing or the removal processing, is performed on each image element **50** included in the X-ray image **201**.

(73) In step **S6**, the image processing device **100** outputs the processed image **22**. The image processing device **100** outputs the processed image **22** to the display device **103** or the server device. In addition, the image processing device **100** stores the processed image **22** in the storage unit **102**.

(74) After displaying the processed image **22** on the display device **103**, the image generation unit **30** may receive the operation input of changing the arithmetic operation parameters. For example, the image generation unit **30** may receive the input of the value of the weight coefficient **23**, or may receive the selection of another preset parameter. For example, the image generation unit **30** may receive the change in the parameters of the image processing in step **S3**. Moreover, the image generation unit **30** may regenerate the processed image **22** using the changed parameters in accordance with the operation input of the user.

(75) (Creation Method of Trained Model)

(76) Next, the creation method of the trained model will be described. The trained model **40** may be created by the processor **101** of the image processing device **100**, but may be performed using a computer for the machine learning (learning device **300**, see FIG. 10).

(77) As shown in FIGS. 9 and 10, the creation method of the trained model according to the present embodiment includes following steps **S11** to **S14**.

(78) (**S11**) The reconstructed image **60** obtained by reconstructing computer tomography (CT) image data **80** into a two-dimensional projective image is generated. The CT image data **80** is an example of “three-dimensional X-ray image data”.

(79) (**S12**) The two-dimensional projection image **61** is generated from the three-dimensional

model of the image element **50**, which is the extraction target, by a simulation.

(80) (S13) The projection image **61** of the image element **50** is superimposed on the reconstructed image **60** to generate a superimposed image **67**.

(81) (S14) The trained model **40** (see FIG. 3) that performs the processing of extracting the image element **50** included in the input image is created by performing the machine learning using the superimposed image **67** as the teacher input data **64** (see FIG. 3) and the reconstructed image **60** or the projection image **61** as the teacher output data **65** (see FIG. 3).

(82) In the present embodiment, the machine learning includes inputting the teacher input data **64** (see FIG. 3) and the teacher output data **65** (see FIG. 3) created for each image element **50** to one learning model LM. The machine learning may be performed on a separate learning model LM for each image element **50**, which is the extraction target.

(83) First, in step S10 of FIG. 10, the CT image data **80** is acquired. The CT image data **80** is three-dimensional image data that reflects a three-dimensional structure of the subject **1** obtained by capturing the CT image of the subject **1**. The CT image data **80** is a three-dimensional aggregate of voxel data including three-dimensional position coordinates, and a CT value at the position coordinates. For the image element of a moving (beating) part, such as the blood vessel, four-dimensional 4D-CT data including temporal change in three-dimensional information may be used. As a result, it is possible to accurately perform learning even in a case in which a target moves with time.

(84) In step S11, the reconstructed image **60** obtained by reconstructing the CT image data **80** into the two-dimensional projective image is generated.

(85) The reconstructed image **60** is a DRR image generated from the CT image data **80**. The DRR image is a simulated X-ray image created as the two-dimensional projective image by virtual fluoroscopic projection simulating geometrical projection conditions of the X-ray irradiation unit **220** and the X-ray detector **230** of the X-ray imaging device **200** as shown in FIG. 2.

(86) Specifically, as shown in FIG. 11, by virtually disposing, in a three-dimensional virtual space, a virtual X-ray tube **91** and a virtual X-ray detector **92** in a predetermined projection direction with respect to the CT image data **80**, a three-dimensional spatial disposition (imaging geometry) of a virtual X-ray imaging system is generated. The disposition of the CT image data **80**, the virtual X-ray tube **91**, and the virtual X-ray detector **92** has the same imaging geometry as the disposition of the actual subject **1**, the X-ray irradiation unit **220**, and the X-ray detector **230** shown in FIG. 2. It should be noted that the imaging geometry means a geometrical disposition relationship between the subject **1**, the X-ray irradiation unit **220**, and the X-ray detector **230** in the three-dimensional space.

(87) Moreover, by adding the total of the CT values in the voxels in which the X-rays emitted from the virtual X-ray tube **91** have passed until reaching the virtual X-ray detector **92**, the pixel value of each pixel in the reconstructed image **60** is calculated. By changing the imaging geometry, it is possible to generate the simulated X-ray image at any projection angle.

(88) In the creation method of the trained model, the learning device **300** generates a plurality of reconstructed images **60** from one three-dimensional data (CT image data **80**). The number of the generated reconstructed images **60** may be, for example, about 100,000. The learning device **300** generates the plurality of reconstructed images **60** by making the parameters, such as a projection angle, a projection coordinate, a parameter for DRR image generation, a contrast, and the edge enhancement, different from each other. The plurality of reconstructed images **60** that are made different from each other may be generated by, for example, an algorithm that randomly changes the parameters described above to generate the reconstructed image **60**.

(89) In the present embodiment, since the superimposed image **67** superimposed on the projection image **61** including the image element **50** is created, the reconstructed image **60** does not have to include the image element **50**, which is the extraction target, or does not have to have an extractable contrast even in a case in which the image element **50** is included.

(90) In addition, in the present specification, the terms “random” and “random number” mean non-regularity and a non-regular sequence (set of numbers), but it does not have to be completely random, but includes pseudo random and a pseudo random number.

(91) In step S12 of FIG. 9, the two-dimensional projection image 61 is generated from the three-dimensional model of the image element 50, which is the extraction target, by a simulation.

(92) The projection image 61 is the two-dimensional image representing the image element 50, which is the target of extraction by the trained model 40. The projection image 61 includes only the image element 50, for example. The learning device 300 acquires the three-dimensional model of the image element 50, which is the extraction target, and generates the projection image 61 from the three-dimensional model by a simulation.

(93) The three-dimensional model is created, for example, by capturing the CT image of a target object including the image element 50, which is the extraction target, and extracting the image element 50 from the obtained CT data. The three-dimensional model may be created using, for example, a CT image database published by a research institution and the like. As the CT image database, for example, as a lung CT image data set, there is the lung image data base consortium and image data base resource initiative (LIDC/IDRI) by the national cancer institute of the United States. In addition to this, a brain CT image data set, a standardized three-dimensional model of the skeleton, or the like may be used. In addition, the device 55 and the clothing 56 may be created using three-dimensional CAD data. The three-dimensional model is, for example, three-dimensional image data including only the image element 50, which is the extraction target.

(94) It should be noted that, it is not necessary to generate all the projection images 61 used for the machine learning from the three-dimensional model by a simulation. The projection image 61 may be created by acquiring the two-dimensional data (X-ray image) actually including the image element 50 to be extracted, and separating and extracting the image element 50 included in the acquired image.

(95) The learning device 300 generates a plurality of two-dimensional projection images 61 by making parameters, such as a projection direction, a translation amount, a rotation amount, a deformation amount, and a contrast, different from each other with respect to a three-dimensional model or two-dimensional data. The plurality of projection images 61 may be generated by an algorithm that performs processing of randomly changing variable parameters, such as the translation amount, the rotation amount, the deformation amount, and the contrast, with respect to the original data (three-dimensional model, two-dimensional data, or other projection images 61).

(96) The learning device 300 generates the plurality of two-dimensional projection images 61 for each type of the image element 50. The learning device 300 generates the plurality of projection images 61 from one original data (three-dimensional data, two-dimensional data, or projection image 61). The number of the projection images 61 generated from one original data may be, for example, about 100,000.

(97) In step S13, the projection image 61 generated in step S12 is superimposed on the reconstructed image 60 generated in step S11 to generate the superimposed image 67 (see FIG. 10). By superimposition, the superimposed image 67 including, in the two-dimensional reconstructed image 60, the projection image 61 of the image element 50 to be extracted is generated. By combining the plurality of reconstructed images 60 and the plurality of projection images 61, a plurality of superimposed images 67 are also generated. One training data 66 (see FIG. 3) includes the superimposed image 67, and any of the reconstructed image 60 and the projection image 61 used to generate the superimposed image 67.

(98) The machine learning is performed in step S14 of FIG. 9. The superimposed image 67 is used as the teacher input data 64 input to the input layer 41 of the learning model LM of FIG. 3. Moreover, the reconstructed image 60 or the projection image 61 is used as the teacher output data 65 input to the output layer 43 of the learning model.

(99) In a case in which the teacher output data 65 is the reconstructed image 60, the learning model

LM is trained to extract the image element **50** from the input image and generate the extraction image **21** not including the image element **50**. In a case in which the teacher output data **65** is the projection image **61**, the learning model LM is trained to extract the image element **50** from the input image and generate the extraction image **21** representing the extraction image element **50**. The extraction image **21** may be an image including only the image element **50**. It should be noted that since the image including only the extraction image element **50** can be generated by generating the processed image **22** not including the image element **50** and performing the subtraction from the input X-ray image **201**, adopting the reconstructed image **60** adopting the projection image **61** as the teacher output data **65** can be considered equivalent from the viewpoint of the image processing.

(100) It should be noted that, it is not necessary to create all the training data **66** used for the machine learning from the superimposed image **67**. The reconstructed image **60** actually including the image element **50** to be extracted may be used as the teacher input data **64**, and the projection image **61** of the image element **50** extracted from the reconstructed image **60** may be used as the teacher output data **65**.

(101) In step **S15** of FIG. **9**, the learning device **300** determines whether or not the machine learning is completed. For example, the learning device **300** determines that the machine learning is completed in a case in which the machine learning is performed a predetermined number of iterations for all the training data **66**. For example, the learning device **300** determines that the machine learning is completed in a case in which a value of an evaluation function for evaluating the performance of the learning model LM is equal to or more than a predetermined value. In a case in which it is determined that the machine learning is not completed, the learning device **300** changes the training data **66** in the step **S16**, and performs the machine learning of the step **S14** using the next training data **66**.

(102) In a case in which the machine learning is completed, the trained learning model LM is stored as the trained model **40** in step **S17**. As a result, the creation of the trained model **40** is completed. The created trained model **40** is recorded via the network or in a non-transitory recording medium and provided to the image processing device **100**.

(103) (Specific Example for Each Image Element)

(104) Next, a specific example of the superimposed image **67** (teacher input data **64**), and the reconstructed image **60** or the projection image **61** (teacher output data **65**) for each image element **50** will be described. In addition, an example of the processed image **22** using the extraction image **21** for each image element **50** will be described.

(105) <Bone Part>

(106) FIG. **12** shows an example in which the image element **50** is the bone **53**. The teacher input data **64** is, for example, the superimposed image **67** of the reconstructed image **60** not including the bone **53**, and the projection image **61** including the bone **53**.

(107) For example, the superimposed image **67** is generated by creating the projection image **61** from the CT image data **80** or the three-dimensional model of the skeleton and superimposing the projection image **61** on the reconstructed image **60** from which the bone **53** is removed. The reconstructed image **60** from which the bone **53** is removed is generated by clipping (fixing) the CT value to zero for the pixels in a range of the CT value of the bone **53**. Since the CT value of the bone part is generally about 200 HU to about 1000 HU, a threshold value need only be set to a predetermined value of about 0 HU to 200 HU.

(108) The teacher output data **65** is the projection image **61** including only the bone **53**. As the teacher output data **65**, the projection image **61** used to generate the superimposed image **67** is used.

(109) It should be noted that, since the CT image data **80** obtained by imaging the subject usually includes the bone part, the reconstructed image **60** generated from the CT image data **80** may be used as the teacher input data **64**. Therefore, it is not always necessary to create the superimposed

image **67**.

(110) In this case, the teacher output data **65** is the reconstructed image **60** including only the bone **53**. For example, the reconstructed image **60** including only the bone **53** is generated by clipping the CT value to zero for the pixels having the CT value less than the CT value of the bone **53** among the reconstructed images **60** including the bone **53**.

(111) Through the machine learning, the learning model LM is trained to generate the extraction image **21** in which the bone **53** is extracted, such as the teacher output data **65**, from the input image, such as the teacher input data **64**.

(112) The image processing device **100** performs the weighting subtraction of the extraction image **21** generated by the trained model **40** on the X-ray image **201** acquired by the X-ray imaging device **200**. As a result, as shown in FIG. **13**, the processed image **22** in which the image element **50** of the bone **53** from the X-ray image **201** is removed is generated. In FIG. **13**, for convenience of description, the removed image element **50** of the bone **53** is shown by a broken line.

(113) <Device>

(114) FIG. **14** shows an example in which the image element **50** is the device **55**. In FIG. **14**, the device **55** is the guidewire.

(115) The teacher input data **64** is the superimposed image **67** including the device **55**. The image element **50** is not included in the reconstructed image **60** generated from the CT image data **80**. The projection image **61** of only the device **55** generated from a three-dimensional model of the device **55** is superimposed on the reconstructed image **60**. As a result, the superimposed image **67** including the device **55** is generated as shown in FIG. **14**.

(116) As described above, a plurality of variations of the projection image **61** of which the shape and the like are changed by a simulation may be created by capturing a two-dimensional X-ray image of the device **55**. The plurality of superimposed images **67** are generated using the plurality of projection images **61**.

(117) In particular, as shown in FIG. **16**, in a case in which the image element **50** is the device **55** having a linear shape or a tubular shape, the projection image **61** of the image element **50** is generated by simulating a shape of the three-dimensional model of the device **55** with a curve generated based on a random coordinate value. FIG. **16** shows examples (A) to (I) in which the projection image **61** of a guidewire **55b** holding a stent **55a** as an example of the device **55** is generated in a random shape by curve simulation.

(118) The guidewires **55b** in the projection images **61** are generated into different shapes by a Bezier curve with the random coordinate value as a base point. The Bezier curve is a K-1 order curve obtained from K control points (K is an integer of 3 or more). A large number of the projection images **61** of the devices **55** having various shapes can be generated by an algorithm that randomly designates the coordinate values of K control points.

(119) The plurality of single projection images **61** of the stent **55a** placed in the body are generated by applying random translation, rotation, deformation, and contrast change to a three-dimensionally modeled pseudo stent.

(120) Returning to FIG. **14**, the teacher output data **65** is the projection image **61** including only the device **55**. As the teacher output data **65**, the projection image **61** used to generate the superimposed image **67** is used.

(121) Through the machine learning, the learning model LM is trained to generate the extraction image **21** in which the device **55** is extracted, such as the teacher output data **65**, from the input image, such as the teacher input data **64**.

(122) The image processing device **100** performs the weighting addition of the extraction image **21** generated by the trained model **40** on the X-ray image **201** acquired by the X-ray imaging device **200**. As a result, as shown in FIG. **15**, the processed image **22** in which the image element **50** of the device **55** is enhanced from the X-ray image **201** is generated. FIG. **15** shows that the device **55** is enhanced by showing the device **55** thicker than that of FIG. **14**. The enhancement processing may

include processing of coloring and displaying the image element **50** of the device **55** by the image processing before the inter-image arithmetic operation shown in FIG. 7, in addition to processing of increasing the pixel value.

(123) <Noise>

(124) FIG. 17 shows an example in which the image element **50** is the noise **57**. The noise **57** is, for example, a random noise, but is shown in FIG. 17 as a set of horizontal dotted lines for convenience of description.

(125) The teacher input data **64** is the superimposed image **67** including the noise **57**. The noise **57** is not included in the reconstructed image **60** generated from the CT image data **80**. The projection image **61** including only the randomly generated noise **57** is superimposed on the reconstructed image **60**. As a result, the superimposed image **67** including the noise **57** is generated as shown in FIG. 17. The noise **57** is created by randomly generating a Gaussian noise that follows a Gaussian distribution or a Poisson noise that follows a Poisson distribution for each projection image **61**.

(126) The teacher output data **65** is the projection image **61** including only the noise **57**. As the teacher output data **65**, the projection image **61** used to generate the superimposed image **67** is used.

(127) Through the machine learning, the learning model LM is trained to generate the extraction image **21** in which the noise **57** is extracted, such as the teacher output data **65**, from the input image, such as the teacher input data **64**.

(128) The image processing device **100** performs the weighting subtraction of the extraction image **21** generated by the trained model **40** on the X-ray image **201** acquired by the X-ray imaging device **200**. As a result, as shown in FIG. 18, the processed image **22** in which the image element **50** of the noise **57** is removed from the X-ray image **201** is generated. The processed image **22** of FIG. 18 shows that the noise **57** is removed from the X-ray image **201** including the noise **57** as in the teacher input data **64** of FIG. 17.

(129) <Blood Vessel>

(130) FIG. 19 shows an example in which the image element **50** is the blood vessel **54**. The blood vessel **54** is a contrast-imaged blood vessel imaged by introducing the contrast medium. FIG. 19 shows an example of a cerebral blood vessel in the head, but another blood vessel may be used. The blood vessel may be, for example, a coronary artery of the heart.

(131) The teacher input data **64** is the superimposed image **67** including the blood vessel **54**. The reconstructed image **60** generated from the CT image data **80** captured without the contrast includes almost no blood vessels **54** (does not have sufficient contrast). The projection image **61** of only the blood vessel **54** generated from a three-dimensional model of the blood vessel **54** is superimposed on the reconstructed image **60**. As a result, the superimposed image **67** including the blood vessel **54** is generated as shown in FIG. 19.

(132) In a case in which the image element **50** is the blood vessel **54**, the projection image **61** of the image element **50** is generated by a simulation that randomly changes a shape of the three-dimensional model of the blood vessel **54**. The blood vessel of the projection image **61** is subjected to random translation, rotation, deformation, contrast change, and the like by a simulation. That is, similar to the device **55** shown in FIG. 16, variations of the projection image **61** of the blood vessel **54** to which a random change is applied are generated. As described above, the projection image **61** of the blood vessel **54** may be created from the CT image data of the contrast-imaged blood vessel captured by the angiography.

(133) The teacher output data **65** is the projection image **61** including only the blood vessel **54**. As the teacher output data **65**, the projection image **61** used to generate the superimposed image **67** is used.

(134) Through the machine learning, the learning model LM is trained to generate the extraction image **21** in which the blood vessel **54** is extracted, such as the teacher output data **65**, from the input image, such as the teacher input data **64**.

(135) The image processing device **100** performs the weighting addition of the extraction image **21** generated by the trained model **40** on the X-ray image **201** acquired by the X-ray imaging device **200**. As a result, as shown in FIG. **20**, the processed image **22** in which the image element **50** of the blood vessel **54** is enhanced from the X-ray image **201** is generated. The processed image **22** of FIG. **20** shows that the blood vessel **54** is enhanced and displayed in the X-ray image **201** including the image element **50** of the blood vessel **54** as in the teacher input data **64** of FIG. **19**.

(136) <Clothing>

(137) FIG. **21** shows an example in which the image element **50** is the clothing **56**. FIG. **21** shows, as an example of the clothing **56**, a button on the clothes and a necklace worn by the subject.

(138) The teacher input data **64** is the superimposed image **67** including the clothing **56**. The clothing **56** is not included in the reconstructed image **60** generated from the CT image data **80**. The projection image **61** of only the clothing **56** generated from a three-dimensional model of the clothing **56** is superimposed on the reconstructed image **60**. As a result, the superimposed image **67** including the clothing **56** is generated as shown in FIG. **21**. A three-dimensional model of the clothing **56** may be created from the CT image of only the clothing **56**, or may be created from, for example, the CAD data. As described above, a two-dimensional X-ray image of the clothing **56** may be captured and used as the projection image **61**. The projection image **61** is subjected to random translation, rotation, deformation, contrast change, and the like by a simulation.

(139) The teacher output data **65** is the projection image **61** including only the clothing **56**. As the teacher output data **65**, the same data as the projection image **61** used to generate the superimposed image **67** is used.

(140) Through the machine learning, the learning model LM is trained to generate the extraction image **21** in which the clothing **56** is extracted, such as the teacher output data **65**, from the input image, such as the teacher input data **64**.

(141) The image processing device **100** performs the weighting subtraction of the extraction image **21** generated by the trained model **40** on the X-ray image **201** acquired by the X-ray imaging device **200**. As a result, as shown in FIG. **22**, the processed image **22** in which the image element **50** of the clothing **56** is removed from the X-ray image **201** is generated. In FIG. **22**, in order to describe the removal, for convenience, a part of the image element **50** of the removed clothing **56** is shown by a broken line.

(142) <Scattered Ray Component>

(143) FIG. **23** shows an example in which the image element **50** is the scattered ray component **58** of the X-rays.

(144) The teacher input data **64** is the superimposed image **67** including the scattered ray component **58**. In the reconstructed image **60** generated from the CT image data **80**, the scattered ray component **58** is not included in a reconstruction arithmetic operation. The projection image **61** of only the scattered ray component **58** generated by the Monte Carlo simulation modeling an imaging environment of the input image is superimposed on the reconstructed image **60**. As a result, the superimposed image **67** including the scattered ray component **58** is generated.

(145) In the Monte Carlo simulation, for example, as shown in FIG. **2** (or FIG. **11**), an imaging environment model **85** obtained by three-dimensionally modeling the imaging environment of the input image (X-ray image **201**) from the X-ray irradiation unit **220** to the X-ray detector **230** is created. Moreover, each of X-ray photons emitted from the X-ray irradiation unit **220** and detected by the X-ray detector **230** is computed (simulated) as a probabilistic event using a random number. That is, in the simulation, the physical properties related to the projection direction of the X-rays, a shape (body shape) of the subject, and the interaction with photons are assumed. Moreover, the interaction, such as absorption and scattering phenomenon, that occurs when the X-ray photon passes through the subject is computed using a random number as the probabilistic event. In the Monte Carlo simulation, a predetermined number of photons is computed, and the projection image **61** formed by the X-ray photons detected by the virtual X-ray detector **230** is generated. The

predetermined number of photons need only be a sufficient number for imaging, for example, about 10 billion.

(146) In the present embodiment, the plurality of the projection images **61** of the image element **50**, which is the scattered ray component **58**, are generated by changing the projection angle over a projection angle range capable of being imaged by the X-ray imaging device **200** in the imaging environment model **85**. For example, in the example shown in FIG. **2**, the projection direction can be changed to the first direction **250** and the second direction **252** by moving the C-arm **240**.

Therefore, as shown in FIG. **25**, the projection image **61** by the Monte Carlo simulation is generated at a plurality of projection angles changed to different angle values over the entire projection angle range of $\pm\alpha$ degrees in the first direction **250** and $\pm\beta$, degrees in the second direction **252**. The projection angle may be changed at equal intervals over the entire projection angle range, or may be a value randomly changed by a predetermined number in the projection angle range. Further, the projection image **61** may be generated at an angle value outside the projection angle range and near a limit of the projection angle ($\pm\alpha$ degrees, $\pm\beta$, degrees).

(147) In addition, in the present embodiment, the plurality of projection images **61** of the image element **50**, which is the scattered ray component **58**, are generated by changing an energy spectrum of the virtual radiation source in the imaging environment model **85**. That is, under a plurality of conditions in which the energy spectrum of the X-rays emitted by the X-ray irradiation unit **220** assumed as the imaging environment model **85** is changed to a different spectrum, the projection image **61** by the Monte Carlo simulation is created.

(148) Generally, as the photon energy of the X-rays is lower, it is more likely to be absorbed in the body of the subject, and it is less likely that the scattered ray component **58** is generated. As the photon energy of the X-rays is higher, it is less likely to be absorbed in the body of the subject, and it is more likely that the scattered ray component **58** is generated. Therefore, for example, the projection image **61** based on a first energy spectrum **111** shown in FIG. **26** and the projection image **61** based on a second energy spectrum **112** shown in FIG. **27** are created. The second energy spectrum **112** is a spectrum that has a relatively higher energy than the first energy spectrum **111**. In FIGS. **26** and **27**, the horizontal axis of the graph indicates the energy [keV] of X-ray photons, and the vertical axis of the graph indicates the relative intensity of the X-rays (that is, the number of detected X-ray photons).

(149) It should be noted that, due to a difference in absorption spectra, a beam hardening phenomenon that is relatively biased toward a high energy side occurs in a process of detecting the energy spectrum of the X-rays emitted to the subject. The reconstructed image **60** generated from the CT image data **80** cannot simulate the change in image quality due to the beam hardening phenomenon, but the projection image **61** by the Monte Carlo simulation can simulate the influence of the beam hardening phenomenon.

(150) FIG. **23** shows, as an example, the projection image **61** of the scattered ray component **58** due to Compton scattering obtained by the Monte Carlo simulation. In the present embodiment, the scattered ray component **58** other than Compton scattering, such as Rayleigh scattering, may be obtained. Further, the scattered ray component **58** due to multiple scattering, in addition to single scattering, may be obtained. The machine learning may be performed such that these various scattered ray components **58** are created as separate projection images **61** and separately extracted, or the machine learning may be performed such that the projection image **61** that collectively displays various scattered ray components **58** are created and various scattered ray components **58** are collectively extracted.

(151) The teacher output data **65** is the projection image **61** including only the scattered ray component **58**. As the teacher output data **65**, the same data as the projection image **61** used to generate the superimposed image **67** is used.

(152) Through the machine learning, the learning model LM is trained to generate the extraction image **21** in which the scattered ray component **58** is extracted, such as the teacher output data **65**,

from the input image, such as the teacher input data **64**.

(153) The image processing device **100** performs the weighting subtraction of the extraction image **21** generated by the trained model **40** on the X-ray image **201** acquired by the X-ray imaging device **200**. As a result, as shown in FIG. **24**, the processed image **22** in which the image element **50** of the scattered ray component **58** is removed from the X-ray image **201** is generated. Since the scattered ray component **58** is a factor that lowers the contrast of the X-ray image, the contrast in the processed image **22** can be improved by removing the scattered ray component **58**. The processed image **22** of FIG. **24** shows that the contrast is improved by removing the scattered ray component **58** from the X-ray image **201** of which the contrast is lowered by the scattered ray component **58** as in the teacher input data **64** of FIG. **23**.

(154) <Collimator Image>

(155) In the present embodiment, a collimator image **68** of which an imaging range is limited by a collimator (not shown) provided in the X-ray imaging device **200** (X-ray irradiation unit **220**) is included in a part of each teacher input data **64** and each teacher output data **65**, which are created. In the collimator image **68**, the image is formed only in a partial region of the image, and a region shielded by the collimator does not include the image information. A part of each teacher input data **64** and each teacher output data **65** includes a plurality of collimator images **68** in which the shape of the irradiation range of the X-rays (that is, image region) and the parameter of the image quality affected by the collimator are randomly made different by a simulation based on the image actually captured by the collimator. The parameters of the image quality affected by the collimator are a degree of transparency (contrast), blur of an edge, a content of the noise, and the like.

(156) The collimator image **68** is generated, for example, by performing the image processing of removing the image part outside the simulated irradiation range and simulating the influence of the collimator on the superimposed image **67**, the reconstructed image **60**, and the projection image **61**. The collimator image **68** may be generated from an image actually captured using the collimator. As a result, it is possible to improve the robustness of the processing of extracting the image element **50** with respect to the change in the irradiation range of the X-rays or the change in the image quality due to the use of the collimator.

(157) As described above, the machine learning for each type of the image element **50** and the generation of the processed image **22** by the image processing device **100** are performed.

(158) In each of the specific examples described above, for convenience of description, each image element **50** and the processed image **22** are described individually. However, in a case in which the processed image **22** is generated by the actual image processing device **100**, the X-ray image **201** input to the image processing device **100** includes a plurality of image elements of the bone **53**, the blood vessel **54**, the device **55**, the clothing **56**, the noise **57**, and the scattered ray component **58**. The image processing device **100** generates the extraction image **21** obtained by extracting the individual image element **50** from the input X-ray image **201** using the trained model **40**, and performs the inter-image arithmetic operation. As a result, the processed image **22** in which the enhancement processing or the removal processing is performed on each of the plurality of image elements **50** is generated.

(159) For example, in the example shown in FIG. **7**, N=6, the first extraction image **21-1** represents the bone **53**, the second extraction image **21-2** represents the device **55**, the third extraction image (**21-3**) represents the noise **57**, the fourth extraction image (**21-4**) represents the blood vessel **54**, the fifth extraction image (**21-5**) represents the clothing **56**, and the sixth extraction image (**21-6**) represents the scattered ray component **58**.

(160) Some examples of the usage scene of the processed image **22** are shown.

(161) For example, the processed image **22** is applied to the X-ray image of the front of the chest of the subject by simple X-ray imaging. In this case, in the processed image **22**, the bone **53**, the noise **57**, the clothing **56**, and the scattered ray component **58** are removed. Due to the removal the bone **53**, the visibility of a region of interest, such as the heart and lungs, is improved. In addition, the

visibility of the entire image is improved by removing the noise **57** and the scattered ray component **58**. Since the image element **50** of the clothing **56** can be removed, the X-ray imaging can be performed in a case in which the subject does not remove the clothes, the accessory, or the like including metal or the like. As a result, useful effects, such as the improvement of the work efficiency and the reduction of waiting time of the subject, can be obtained in a case in which the X-ray imaging of a large number of subjects is continuously performed, such as in mass examination.

(162) In addition, for example, the processed image **22** is applied to an X-ray fluoroscopic image in X-ray interventional radiology (IVR), such as catheter treatment using an X-ray blood vessel imaging device. In this case, in the processed image **22**, the bone **53**, the noise **57**, and the scattered ray component **58** are removed. In the processed image **22**, the device **55**, such as the catheter, the guidewire, or the stent, and the blood vessel **54** are enhanced. Due to the removal of the bone **53**, the noise **57**, and the scattered ray component **58**, the visibility of the fluoroscopic image is improved. Due to the enhancement of the device **55** and the blood vessel **54**, the visibility of the region of interest or the device being operated in catheter treatment is improved.

Effect of Present Embodiment

(163) In the present embodiment, the following effects can be obtained.

(164) With the creation method of the trained model **40** according to the present embodiment, the superimposed image **67** obtained by superimposing the reconstructed image **60** obtained by reconstructing the CT image data **80** into the two-dimensional projective image on the two-dimensional projection image **61** generated from the three-dimensional model of the image element **50**, which is the extraction target, by a simulation is used as the teacher input data **64**, and the reconstructed image **60** or the projection image **61** is used as the teacher output data **65**. As a result, even in a case in which the CT image data **80** does not include the image element **50**, which is the extraction target, the machine learning can be performed using the image element **50**, which is the extraction target, generated by a simulation. That is, the teacher data can be prepared without actually preparing the CT image data **80** including the image element **50** to be extracted. In addition, since the projection image **61** of the image element **50**, which is the extraction target, is generated by the simulation, the teacher data can be prepared even for the image element **50** that is included in the CT image data **80**, but is difficult to be separated and extracted. As a result, it is possible to efficiently create the trained model **40** for performing the image processing on various image elements **50** and even on the plurality of image elements **50**.

(165) With the image generation method and the image processing device **100** according to the present embodiment, the plurality of image elements **50** are separately extracted from the X-ray image **201** using the trained model **40** in which the processing of extracting the specific image element **50** from the input image has been learned, and the processed image **22** is generated by performing the inter-image arithmetic operation using the plurality of extraction images **21** extracted for the respective image elements **50**, and the X-ray image **201**. As a result, various image elements **50** can be separately extracted as the extraction image **21** from the input X-ray image **201**, and each extraction image **21** can be freely added or subtracted from the X-ray image **201** in accordance with the type of the extraction image element **50**. As a result, the image processing can be performed on various image elements **50** and even on the plurality of image elements **50**.

(166) In addition, in the example of the embodiment described above, a further effect can be obtained by the following configuration.

(167) That is, in the present embodiment, a plurality of the superimposed images **67** are created for each of the plurality of image elements **50** different from each other, and the plurality of image elements **50** include the first element **51**, which is the biological tissue, and the second element **52**, which is the non-biological tissue. With this configuration, it is possible to create the trained model **40** that can perform, in a complex manner, the image processing with respect to the image element **50** of the biological tissue, such as the bone **53** or the blood vessel **54**, and the image processing

with respect to the image element **50** of the non-biological tissue, such as the device **55** introduced into the body or the clothing **56** worn by the subject. By using such a trained model **40** in the image processing device **100**, it is possible to perform, in a complex manner, the image processing with respect to the image element **50** of the biological tissue and the image processing with respect to the image element **50** of the non-biological tissue.

(168) In addition, in the present embodiment, the plurality of the superimposed images **67** are created for each of the plurality of image elements **50** different from each other, and the plurality of image elements **50** include at least a plurality of image elements of the bone **53**, the blood vessel **54**, the device **55** introduced into the body, the clothing **56**, the noise **57**, and the scattered ray component **58** of the X-rays. With this configuration, it is possible to create the trained model **40** that can perform the image processing on various image elements **50** in accordance with various usage scenes of the X-ray image **201** in a complex manner. By using such a trained model **40** in the image processing device **100**, it is possible to perform the image processing on various image elements **50** in accordance with various usage scenes of the X-ray image **201** in a complex manner.

(169) In addition, in the present embodiment, the image element **50** includes the device **55** having a linear shape or a tubular shape, the projection image **61** of the image element **50** is generated by simulating a shape of the three-dimensional model of the device **55** with a curve generated based on a random coordinate value. With this configuration, the teacher data for learning the image element **50** of the device **55**, which is long and bends into various shapes, such as the guidewire and the catheter, can be generated in a large amount in various shapes by a simulation. As a result, efficient machine learning can be performed without actually preparing a large amount of the three-dimensional CT data in a state in which the device **55** is disposed in the body of the subject.

(170) In addition, in the present embodiment, the image element **50** includes the blood vessel **54**, the projection image **61** of the image element **50** is generated by a simulation that randomly changes a shape of the three-dimensional model of the blood vessel **54**. With this configuration, the teacher data for learning the image element **50** of the blood vessel **54** that is long and bends into a complicated shape can be generated by a simulation in a large amount and including various individual differences. As a result, efficient machine learning can be performed without preparing a large amount of the three-dimensional CT data of various subjects.

(171) In addition, in the present embodiment, the image element **50** includes the scattered ray component **58** of the X-rays, and the projection image **61** of the image element **50** is generated by the Monte Carlo simulation modeling the imaging environment of the input image. With this configuration, the projection image **61** of the scattered ray component **58** of the X-rays, which is difficult to be separated and extracted from the actual three-dimensional CT data or the two-dimensional X-ray image **201**, can be generated by the Monte Carlo simulation. The image processing of removing the scattered ray component **58** from the X-ray image **201** obtained by actually imaging the subject by creating the trained model **40** that extracts the scattered ray component **58** using the projection image **61** generated in this way can be realized without performing complicated high-load computing processing, such as the Monte Carlo simulation. Therefore, for example, it is possible to effectively improve the contrast in the vicinity of the abdomen or the bone part, which is greatly affected by the scattered ray component **58**. In addition, for example, in the X-ray imaging, it is possible to perform the imaging without using an X-ray absorption grid used to reduce the influence of the scattered ray component **58**, and to remove the scattered ray component **58** by the image processing. In that case, since an X-ray dose can be reduced by an amount that the X-ray absorption grid is not used, an exposure dose of the subject can be effectively reduced.

(172) In addition, in the present embodiment, the plurality of the projection images **61** of the image element **50** are generated by changing the projection angle over the projection angle range ($\pm\alpha$, $\pm\beta$) capable of being imaged by the X-ray imaging device **200** in the imaging environment model **85**. With this configuration, it is possible to create a highly versatile trained model **40** that can

effectively extract the scattered ray component **58** even in a case in which the X-ray imaging is performed at various projection angles in the X-ray interventional radiology or the X-ray imaging is performed while changing the projection angles, in addition to the imaging is performed only from a specific projection direction (front of the chest, side of the chest, or the like), such as the simple X-ray imaging.

(173) In addition, in the present embodiment, the plurality of projection images **61** of the image element **50** are generated by changing an energy spectrum of the virtual radiation source in the imaging environment model **85**. In an actual medical field, the X-ray imaging is performed under various imaging conditions having different energy spectra in accordance with the imaging part or the like. However, with the configuration described above, it is possible to create a highly versatile trained model **40** that can effectively extract the scattered ray component **58** even in a case in which such an X-ray image **201** is captured with various energy spectra.

(174) In addition, in the present embodiment, the machine learning includes inputting the teacher input data **64** and the teacher output data **65** created for each image element **50** to one learning model LM, and the trained model **40** is configured to extract the plurality of image elements **50** from the input image without duplication, and output the extracted plurality of image elements **50** and the residual image element **59** remaining after extraction, respectively. With this configuration, it is possible to provide the trained model **40** that can extract the image element **50** such that the original input image is returned in a case in which all the extracted plurality of image elements **50** and the residual image element **59** are added. That is, even in a case in which the plurality of image elements **50** are extracted from one X-ray image **201**, a loss of the image information included in the X-ray image **201** or unintended increase in the image information does not occur. Since a doctor or the like who makes a diagnosis using the X-ray image **201** finds the basis of the diagnosis in the image information included in the original image even in a case in which various image processing for improving the visibility are performed, it is possible to provide the reliable image for the doctor or the like even in a case in which the complex image processing is performed, with the trained model **40** that can extract the image element **50** such that the loss of the image information does not occur.

(175) In addition, in the present embodiment, the image processing is performed separately on a part or all of the plurality of extraction images **21**, and the processed image **22** is generated by the inter-image arithmetic operation between the plurality of extraction images **21** after the image processing, and the X-ray image **201**. With this configuration, by utilizing the fact that the plurality of image elements **50** can be separately extracted using the trained model **40**, the image processing, such as the correction processing or the complementation processing, can be independently performed on the extraction image **21** of each extraction image element **50**. Here, for example, in a case in which an attempt is made to perform the image processing that acts only on the specific image element **50** on the X-ray image **201** before extraction, the image processing algorithm is complicated and heavy, and the image elements **50** other than the specific image element **50** may be adversely affected. On the other hand, in a case in which the image processing is performed on the extraction image **21**, since there is no other image element **50**, high image processing accuracy can be obtained by simply performing simple filtering processing on the entire extraction image **21**. Moreover, by performing the inter-image arithmetic operation using the extraction image **21** after the image processing, the processed image **22** in which the highly accurate image processing limited to the individual image element **50** is performed in addition to simply enhancing or removing the plurality of image elements **50** can be easily generated.

(176) In addition, in the present embodiment, the inter-image arithmetic operation includes the weighting addition or the weighting subtraction of the individual extraction image **21** with respect to the X-ray image **201**. With this configuration, it is possible to obtain the processed image **22** in which the extracted plurality of image elements **50** are individually enhanced or removed one by one. In addition, since the degree of enhancement or the degree of removal of the individual image

element **50** can be appropriately set by adjusting the weight coefficient **23**, it is possible to generate the processed image **22** that can obtain high visibility for the required image element **50** in accordance with various usage scenes of the X-ray image **201**.

Modification Example

(177) It should be noted that the embodiment disclosed this time is an exemplary example in all respects and are not considered to be restrictive. The scope of the present invention is shown by the scope of claims, not the description of the above embodiment, and further includes all changes (modification examples) within the meaning and scope equivalent to the scope of claims.

(178) For example, in the embodiment described above, an example is shown in which the device that performs the machine learning (learning device **300**) and the image processing device **100** are separate devices, but the present invention is not limited thereto. In the present invention, the machine learning may be performed in the image processing device. In addition, the learning device **300** may be configured by a server computer provided on the cloud.

(179) In addition, in the embodiment described above (see FIGS. **5** to **7**), an example in which the plurality of image elements **50** are extracted by one trained model **40** is shown, but the present invention is not limited thereto. In the present invention, the plurality of image elements **50** may be extracted by a plurality of trained models. For example, as shown in FIG. **29**, one trained model **40** may be provided for each image element **50** to be extracted. In FIG. **29**, a trained model **40-1**, a trained model **40-2**, . . . , and a trained model **40-N** are provided. One trained model **40** extracts one (one type of) image element **50**. In addition, a plurality of trained models **40** created to extract the plurality of image elements **50** may be provided.

(180) In addition, in the embodiment described above, an example is shown in which the plurality of image elements **50** include at least a plurality of image elements of the bone **53**, the blood vessel **54**, the device **55**, the clothing **56**, the noise **57**, and the scattered ray component **58**, but the present invention is not limited thereto. The image element **50** may include the image elements other than the bone **53**, the blood vessel **54**, the device **55**, the clothing **56**, the noise **57**, and the scattered ray component **58**. The image element **50** may be a specific structural part, such as a specific organ in the body. In addition, for example, the image element **50** of the bone at the specific part among the bones **53** or the image element **50** of the blood vessel at the specific part among the blood vessels **54** may be separately extracted from other bones or other blood vessels. The image element **50** does not have to include the bone **53**, the blood vessel **54**, the device **55**, the clothing **56**, the noise **57**, and the scattered ray component **58**.

(181) In addition, in the embodiment described above, an example is shown in which the removal processing is performed on the bone **53** and the enhancement processing is performed the blood vessel **54** and the device **55** by the inter-image arithmetic operation, but the present invention is not limited thereto. The enhancement processing may be performed on the bone **53**, or the removal processing may be performed on one or both of the blood vessel **54** and the device **55**.

(182) In addition, in the embodiment described above, two examples of the weighting addition and the weighting subtraction are shown as examples of the inter-image arithmetic operation, but the present invention is not limited thereto. The inter-image arithmetic operation may be addition or subtraction without the weight coefficient. The enhancement processing of the image element **50** may be performed by multiplication with the weight coefficient or without the weight coefficient. The removal processing of the image element **50** may be performed by division with the weight coefficient or without the weight coefficient.

ASPECT

(183) It will be understood by those skilled in the art that the exemplary embodiment described above is a specific example of the following aspects.

(184) (Item 1)

(185) A creation method of a trained model, the method including generating a reconstructed image obtained by reconstructing three-dimensional X-ray image data into a two-dimensional projective

image, generating a two-dimensional projection image from a three-dimensional model of an image element, which is an extraction target, by a simulation, superimposing the projection image of the image element on the reconstructed image to generate a superimposed image, and creating a trained model that performs processing of extracting the image element included in an input image, by performing machine learning using the superimposed image as teacher input data and the reconstructed image or the projection image as teacher output data.

(186) (Item 2)

(187) The creation method of a trained model according to item 1, in which a plurality of the superimposed images are created for each of a plurality of image elements different from each other, and the plurality of image elements include a first element, which is a biological tissue, and a second element, which is a non-biological tissue.

(188) (Item 3)

(189) The creation method of a trained model according to item 1, in which a plurality of the superimposed images are created for each of a plurality of image elements different from each other, and the plurality of image elements include at least a plurality of image elements of a bone, a blood vessel, a device introduced into a body, clothing, a noise, and a scattered ray component of X-rays.

(190) (Item 4)

(191) The creation method of a trained model according to item 1, in which the image element includes a device having a linear shape or a tubular shape, and the projection image of the image element is generated by simulating a shape of a three-dimensional model of the device with a curve generated based on a random coordinate value.

(192) (Item 5)

(193) The creation method of a trained model according to item 1, in which the image element includes a blood vessel, and the projection image of the image element is generated by a simulation that randomly changes a shape of a three-dimensional model of the blood vessel.

(194) (Item 6)

(195) The creation method of a trained model according to item 1, in which the image element includes a scattered ray component of X-rays, and the projection image of the image element is generated by a Monte Carlo simulation that models an imaging environment of the input image.

(196) (Item 7)

(197) The creation method of a trained model according to item 6, in which a plurality of the projection images of the image element are generated by changing a projection angle over a projection angle range capable of being imaged by an X-ray imaging device in an imaging environment model.

(198) (Item 8)

(199) The creation method of a trained model according to item 6, in which a plurality of the projection images of the image element are generated by changing an energy spectrum of a virtual radiation source in an imaging environment model.

(200) (Item 9)

(201) The creation method of a trained model according to item 1, in which the machine learning includes inputting the teacher input data and the teacher output data created for each image element to one learning model, and the trained model is configured to extract the plurality of image elements from the input image without duplication, and output the extracted plurality of image elements and a residual image element remaining after extraction, respectively.

(202) (Item 10)

(203) An image generation method including separately extracting a plurality of image elements from an X-ray image using a trained model in which processing of extracting a specific image element from an input image has been learned, and generating a processed image in which image processing is performed on each image element included in the X-ray image, by performing an

inter-image arithmetic operation using a plurality of extraction images extracted for the respective image elements, and the X-ray image.

(204) (Item 11)

(205) The image generation method according to item 10, in which the image processing includes enhancement processing or removal processing.

(206) (Item 12)

(207) The image generation method according to item 10, in which the plurality of image elements include a first element, which is a biological tissue, and a second element, which is a non-biological tissue.

(208) (Item 13)

(209) The image generation method according to item 10, in which the plurality of image elements include at least a plurality of image elements of a bone, a blood vessel, a device introduced into a body, clothing, a noise, and a scattered ray component of X-rays.

(210) (Item 14)

(211) The image generation method according to item 10, in which the image processing is performed separately on a part or all of the plurality of extraction images, and the processed image is generated by the inter-image arithmetic operation between the plurality of extraction images after the image processing, and the X-ray image.

(212) (Item 15)

(213) The image generation method according to item 10, in which the inter-image arithmetic operation includes weighting addition or weighting subtraction of an individual extraction image with respect to the X-ray image.

(214) (Item 16)

(215) The image generation method according to item 10, in which the trained model is configured to extract the plurality of image elements from the input image without duplication, and output the extracted plurality of image elements and a residual image element remaining after extraction, respectively.

(216) (Item 17)

(217) The image generation method according to item 10, in which the trained model is created in advance through machine learning using a reconstructed image obtained by reconstructing three-dimensional image data into a two-dimensional projective image, and a projection image created from a three-dimensional model of the image element by a simulation.

(218) (Item 18)

(219) An image processing device including an image acquisition unit that acquires an X-ray image, an extraction processing unit that separately extracts a plurality of image elements from the X-ray image using a trained model in which processing of extracting a specific image element from an input image has been learned, and an image generation unit that generates a processed image in which image processing is performed on each image element included in the X-ray image, by performing an inter-image arithmetic operation using a plurality of extraction images extracted for the respective image elements, and the X-ray image.

REFERENCE SIGNS LIST

(220) **10**: Image acquisition unit **20**: Extraction processing unit **21** (**21-1**, **21-2**, **21-N**): extraction image **22**: Processed image **30**: Image generation unit **40** (**40-1**, **40-2**, **40-N**): Trained model **50**: Image element **51**: First element **52**: Second element **53**: bone **54**: Blood vessel **55**: device **56**: Clothing **57**: Noise **58**: Scattered ray component **59**: Residual image element **60**: Reconstructed image **61**: Projection image **64**: Teacher input data **65**: Teacher output data **67**: Superimposed image **80**: CT image data (three-dimensional X-ray image data) **85**: Imaging environment model **100**: Image processing device **111**: First energy spectrum (energy spectrum) **112**: Second energy spectrum (energy spectrum) **200**: X-ray imaging device **201**: X-ray image LM: learning model

Claims

1. A creation method of a trained model, the method comprising: generating a reconstructed image obtained by reconstructing three-dimensional X-ray image data into a two-dimensional projective image; generating a two-dimensional projection image from a three-dimensional model of an image element, which is an extraction target, by a simulation; superimposing the projection image of the image element on the reconstructed image to generate a superimposed image; and creating a trained model that performs processing of extracting a plurality of the image element included in an input image and generating a plurality of extraction images which the plurality of the image element are reflected therein, by performing machine learning using the superimposed image as teacher input data and the reconstructed image or the projection image as teacher output data.
2. The creation method of a trained model according to claim 1, wherein a plurality of the superimposed images are created for each of a plurality of image elements different from each other, and the plurality of image elements include a first element, which is a biological tissue, and a second element, which is a non-biological tissue, or at least a plurality of image elements of a bone, a blood vessel, a device introduced into a body, clothing, a noise, and a scattered ray component of X-rays.
3. The creation method of a trained model according to claim 1, wherein the image element includes a device having a linear shape or a tubular shape, and the projection image of the image element is generated by simulating a shape of a three-dimensional model of the device with a curve generated based on a random coordinate value.
4. The creation method of a trained model according to claim 1, wherein the image element includes a blood vessel, and the projection image of the image element is generated by a simulation that randomly changes a shape of a three-dimensional model of the blood vessel.
5. The creation method of a trained model according to claim 1, wherein the image element includes a scattered ray component of X-rays, and the projection image of the image element is generated by a Monte Carlo simulation that models an imaging environment of the input image.
6. The creation method of a trained model according to claim 5, wherein a plurality of the projection images of the image element are generated by changing a projection angle over a projection angle range capable of being imaged by an X-ray imaging device in an imaging environment model, or by changing an energy spectrum of a virtual radiation source in an imaging environment model.
7. The creation method of a trained model according to claim 1, wherein the machine learning includes inputting the teacher input data and the teacher output data created for each image element to one learning model, and the trained model is configured to extract the plurality of image elements from the input image without duplication, and output the extracted plurality of image elements and a residual image element remaining after extraction, respectively.
8. An image generation method comprising: separately generating a plurality of extraction images by a plurality of image elements from an X-ray image using a trained model in which processing of extracting an image element from an input image has been learned; and generating a processed image in which image processing is performed on each image element included in the X-ray image, by weighting each of the plurality of extraction images and adding or subtracting the weighted plurality of extraction images to or from the X-ray image.
9. The image generation method according to claim 8, wherein the image processing includes enhancement processing or removal processing.
10. The image generation method according to claim 8, wherein the plurality of image elements include a first element, which is a biological tissue, and a second element, which is a non-biological tissue, or at least a plurality of image elements of a bone, a blood vessel, a device introduced into a body, clothing, a noise, and a scattered ray component of X-rays.

11. The image generation method according to claim 8, wherein the image processing is performed separately on a part or all of the plurality of extraction images, and the processed image is generated by the plurality of extraction images after the image processing, and the X-ray image.
 12. The image generation method according to claim 8, wherein the trained model is configured to extract the plurality of image elements from the input image without duplication, and output the extracted plurality of image elements and a residual image element remaining after extraction, respectively.
 13. The image generation method according to claim 8, wherein the trained model is created in advance through machine learning using a reconstructed image obtained by reconstructing three-dimensional image data into a two-dimensional projective image, and a projection image created from a three-dimensional model of the image element by a simulation.
 14. An image processing device comprising: an image acquisition unit that acquires an X-ray image; an extraction processing unit that separately generates a plurality of extraction images by a plurality of image elements from the X-ray image using a trained model in which processing of extracting an image element from an input image has been learned; and an image generation unit that generates a processed image in which image processing is performed on each image element included in the X-ray image, by weighting each of the plurality of extraction images and adding or subtracting the weighted plurality of extraction images to or from the X-ray image.
 15. A creation method of a trained model, the method comprising: generating a reconstructed image obtained by reconstructing three-dimensional X-ray image data into a two-dimensional projective image; generating a two-dimensional projection image by simulating a shape of a three-dimensional model of an image element of a linear or tubular shape curved device, which is an extraction target, with a curve; superimposing the projection image of the image element on the reconstructed image to generate a superimposed image; and creating a trained model used in a process of extracting the image element included in an input image, by performing machine learning using the superimposed image as teacher input data and the reconstructed image or the projection image as teacher output data.
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